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DETERMINING A WEIGHT SET TO PERSONALIZE A RULE THAT GOVERNS A MANNER OF A PERFORMANCE OF A CHANGE OF A MOVEMENT OF AN AUTOMATED VEHICLE

Abstract

A system for determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle can determine the weight from responses, from a human subject, to questions about preferences about the manner of the performance. The questions can be selected, using a greedy technique, from a question set. The weight can be applied to the rule to incorporate the preference of the human subject about the manner of the performance. The preference can be with respect to one or more of: (1) a degree of comfort of the human subject during the performance, (2) a degree of vehicle performance of the automated vehicle during the performance, or (3) the like. Such a rule can increase a likelihood that a human driver (i.e., the human subject) will accept having the automated vehicle operated under control of an automated driving system.

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Background/Summary

[0001] This application claims the benefit of U.S. Provisional Application No. 63/562,910, filed Mar. 8, 2024, which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The disclosed technologies are directed to determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle.

BACKGROUND

[0003] Driving automation technology can include, for example, an autonomous motion technology system. An autonomous motion technology system can be used to cause a vehicle to move independently in an environment with one or more other vehicles and/or objects. Such an autonomous motion technology system can be arranged to perform functions in stages. Such stages can include, for example, an information acquisition stage, a perception stage, and a control stage. The information acquisition stage can include technologies through which the vehicle can obtain information about the one or more other vehicles and/or objects in the environment of the vehicle and/or information about a location and/or a movement of the vehicle. The perception stage can perform functions on information from the information acquisition stage to produce information that facilitates a better understanding of the environment of the vehicle. The control stage can perform functions on information from the perception stage to produce: (1) a planned trajectory for the vehicle and (2) a control signal to cause the vehicle to move according to the planned trajectory.

SUMMARY

[0004] In an embodiment, a system for determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle can include a processor and a memory. The memory can store a weight determination module, a perception module, a rule production module, and an implementation module. The weight determination module can include instructions that, when executed by the processor, cause the processor to determine from responses, from a human subject, to questions about preferences about the manner of the performance of the change of the movement of the automated vehicle, the weight for the rule. The questions can be selected, using a greedy technique, from a question set. The perception module can include instructions that, when executed by the processor, cause the processor to determine an existence of a reason to cause the change of the movement of the automated vehicle. The rule production module can include instructions that, when executed by the processor, cause the processor to produce the rule, with the weight applied, that governs the manner of the performance of the change of the movement of the automated vehicle. The rule can include: (1) a first subrule for a preference with respect to the manner of the performance of the change of the movement of the automated vehicle and (2) a second subrule for a safety of the manner of the performance of the change of the movement of the automated vehicle. The implementation module can include instructions that, when executed by the processor, cause the change of the movement of the automated vehicle.

[0005] In another embodiment, a method for determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle can include determining, by a processor and from responses, from a human subject, to questions about preferences about the manner of the performance of the change of the movement of the automated vehicle, the weight for the rule. The questions can be selected, using a greedy technique, from a question set. The method can include determining, by the processor, an existence of a reason to cause the change of the movement of the automated vehicle. The method can include producing, by the processor, the rule, with the weight applied, that governs the manner of the performance of the change of the movement of the automated vehicle. The rule can include: (1) the first subrule for a preference with respect to the manner of the performance of the change of the movement of the automated vehicle and (2) a second subrule for a safety of the manner of the performance of the change of the movement of the automated vehicle. The method can include causing, by the processor, the change of the movement of the automated vehicle.

[0006] In another embodiment, a non-transitory computer-readable medium for determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle can include instructions that, when executed by one or more processors, cause the one or more processors to determine, from responses, from a human subject, to questions about preferences about the manner of the performance of the change of the movement of the automated vehicle, the weight for the rule that governs the manner of the performance of the change of the movement of the automated vehicle. The questions can be selected, using a greedy technique, from a question set. The non-transitory computer-readable medium can include instructions that, when executed by the one or more processors, cause the one or more processors to determine an existence of a reason to cause the change of the movement of the automated vehicle. The non-transitory computer-readable medium can include instructions that, when executed by the one or more processors, cause the one or more processors to produce the rule, with the weight applied, that governs the manner of the performance of the change of the movement of the automated vehicle. The rule can include: (1) a first subrule for a preference with respect to the manner of the performance of the change of the movement of the automated vehicle and (2) a second subrule for a safety of the manner of the performance of the change of the movement of the automated vehicle. The non-transitory computer-readable medium can include instructions that, when executed by the one or more processors, cause the one or more processors to cause the change of the movement of the automated vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate various systems, methods, and other embodiments of the disclosure. It will be appreciated that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one embodiment of the boundaries. In some embodiments, one element may be designed as multiple elements or multiple elements may be designed as one element. In some embodiments, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale.

[0008] FIG. **1** includes a diagram that illustrates an example of an environment for determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle, according to the disclosed technologies.

[0009] FIG. **2** includes a diagram that illustrates an example of an automated vehicle configured to apply a weight for a rule that governs a manner of a performance of a change of a movement of the automated vehicle, according to the disclosed technologies.

[0010] FIG. **3** includes a block diagram that illustrates a first example of a system for determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle, according to the disclosed technologies.

[0011] FIG. **4** includes a block diagram that illustrates a second example of the system for determining the weight for the rule that governs the manner of the performance of the change of the movement of the automated vehicle, according to the disclosed technologies.

[0012] FIG. **5** includes graphs that illustrate examples of speed as a function of a specific distance to a point at which an automated vehicle is to stop that can be used for a first example of an employment of the disclosed technologies.

[0013] FIG. **6** includes graphs that illustrate examples of lateral distance as a function of longitudinal distance that can be used for a second example of an employment of the disclosed technologies.

[0014] FIG. **7** includes a flow diagram that illustrates an example of a method that is associated with determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle, according to the disclosed technologies.

[0015] FIG. **8** includes a block diagram that illustrates an example of elements disposed on a vehicle, according to the disclosed technologies.

DETAILED DESCRIPTION

[0016] The disclosed technologies are directed to directed to determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle. For example, the automated vehicle can be an autonomous vehicle. For example, the rule can be expressed using weighted signal temporal logic. The weight for the rule can be determined, from responses, from a human subject, to questions about preferences about the manner of the performance of the change of the movement of the automated vehicle. The questions can be selected, using a greedy technique, from a question set. For example, the greedy technique can be a problem-solving heuristic in which, at each stage of the problem-solving heuristic, a choice of the problem-solving heuristic is a locally optimal choice. That is, for example, because at each stage the choice produced by the greedy technique is the locally optimal choice, the greedy technique may not produce an optimal solution, but can produce, in a reasonable amount of time, locally optimal solutions that approximate a globally optimal solution. An existence of a reason to cause the change of the movement of the automated vehicle can be determined. For example, the existence of the reason to cause the change of the movement of the automated vehicle can be determined after the weight for the rule has been determined. For example, a perception stage of an autonomous motion technology system can determine, from information from an information acquisition stage (e.g., a sensor, a communications device, etc.) of the autonomous motion technology system, the existence of the reason to cause the change of the movement of the automated vehicle. The rule, with the weight applied, that governs the change of the movement of the automated vehicle can be produced. For example, a control stage of the autonomous motion technology system can produce the rule with the weight applied. The rule can include a first subrule and a second subrule. The first subrule can be for a preference with respect to the manner of the performance of the change of the movement of the automated vehicle. For example, the preference can be with respect to one or more of: (1) a degree of comfort of the human subject during the performance of the change of the movement of the automated vehicle, (2) a degree of vehicle performance of the automated vehicle during the performance of the change of the movement of the automated vehicle, or (3) the like. The second subrule can be for a safety of the manner of the performance of the change of the movement of the automated vehicle. The change of the movement of the automated vehicle can be caused to occur. For example, the control stage can cause the change of the movement of the automated vehicle to occur.

[0017] The Society of Automotive Engineers (SAE) International has specified various levels of driving automation. Specifically, Standard J3016, Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles, issued by the SAE International on Jan. 16, 2014, and most recently revised on Apr. 30, 2021, defines six levels of driving automation. These six levels include: (1) level 0, no automation, in which all aspects of dynamic driving tasks are performed by a human driver; (2) level 1, driver assistance, in which a driver assistance system, if selected, can execute, using information about the driving environment, either steering or acceleration/deceleration tasks, but all remaining driving dynamic tasks are performed by a human driver; (3) level 2, partial automation, in which one or more driver assistance systems, if selected, can execute, using information about the driving environment, both steering and acceleration/deceleration tasks, but all remaining driving dynamic tasks are performed by a human driver; (4) level 3, conditional automation, in which an automated driving system, if selected, can execute all aspects of dynamic driving tasks with an expectation that a human driver will respond appropriately to a request to intervene; (5) level 4, high automation, in which an automated driving system, if selected, can execute all aspects of dynamic driving tasks even if a human driver does not respond appropriately to a request to intervene; and (6) level 5, full automation, in which an automated driving system can execute all aspects of dynamic driving tasks under all roadway and environmental conditions that can be managed by a human driver.
[0018] Efforts to automate vehicles can be pursued for a variety of reasons. Among such reasons can be a desire to reduce an amount of fuel consumed by a vehicle. For example, it is estimated that by 2050 that autonomous vehicles may reduce consumption of fuel by passenger vehicles by 44 percent, and by trucks by 18 percent. However, realization of such a benefit can depend upon a degree of satisfaction, of a human driver of an automated vehicle, with a manner in which the automated vehicle is operated under a control of an automated driving system. If the human driver is dissatisfied with the manner in which the automated vehicle is operated under the control of the automated driving system, then the human driver may cause control of the automated vehicle to be transferred from the automated driving system to the human driver.
[0019] For example, if the human driver is dissatisfied with: (1) a degree of comfort of the human driver during a performance of the change of the movement of the automated vehicle, (2) a degree of vehicle performance of the automated vehicle during the performance of the change of the movement of the automated vehicle, or (3) the like, then the human driver may cause control of the automated vehicle to be transferred from the automated driving system to the human driver. Such a transfer of control of the automated vehicle can undermine, for example, the advantage in reduced consumption of fuel intended to be realized when the automated vehicle is under the driving automation technology system or the likelihood that the human driver will accept having the automated vehicle operated under the control of the automated driving system.
[0020] By having a rule that can govern a manner of a performance of a change of a movement of an automated vehicle include: (1) a first subrule and (2) a second subrule in which: (1) the first subrule is for a preference with respect to the manner of the preference of the change of the movement of the automated vehicle, (2) the second subrule is for a safety of the manner of the performance of the change of the movement of the automated vehicle, and (3) a weight, determined from responses, from the human subject, to questions about preferences about the manner of the performance of the change of the movement of the automated vehicle, is applied to the rule, the rule produced by the disclosed technologies can incorporate the preference of the human subject about the manner of the performance of the change of the movement of the automated vehicle while ensuring that the manner of the performance of the change of the movement of the automated vehicle is done safely. Such a rule can increase the likelihood that the human driver (i.e., the human subject) will accept having the automated vehicle operated under the control of the automated driving system.
[0021] FIG. **1** includes a diagram **100** that illustrates an example of an environment for determining

a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle, according to the disclosed technologies. For example, the diagram **100** can include a first road **102** (disposed along a line of latitude) and a second road **104** (disposed along a line of longitude). For example, the environment **100** can include a road junction **106** (e.g., an intersection) of the first road **102** and the second road **104**. For example, the first road **102** can include a westbound lane **108** and an eastbound lane **110**. For example, the second road **104** can include a southbound lane **112** and a northbound lane **114**. For example, the diagram **100** can include a stop sign **116** at a southeast corner of the road junction **106**. For example, the diagram **100** can include a crosswalk **118** disposed across the second road **104** south of the road junction **106**. For example, the diagram **100** can include: (1) a first vehicle **120** in the eastbound lane **110** west of the road junction **106**, (2) a second vehicle **122** in the eastbound lane **110** west of the road junction **106**, but east of the first vehicle **120**, and (3) a third vehicle **124** in the northbound lane **114** south of the road junction **106**. For example, the first vehicle **120** can have a communications device **126**. For example, the third vehicle **124** can have a communications device **128**. For example, the diagram **100** can include a computing device **130**. For example, the computing device **130** can have a communications device **132**. For example, the diagram **100** can include a pedestrian **134** located in the crosswalk **118** in the northbound lane **114**.

[0022] FIG. **2** includes a diagram **200** that illustrates an example of an automated vehicle **202** configured to apply a weight for a rule that governs a manner of a performance of a change of a movement of the automated vehicle **202**, according to the disclosed technologies. For example, the automated vehicle **202** can include one or more wheels **204**, a steering linkage **206** connected to the one or more wheels **204**, one or more brakes **208** for the one or more wheels **204**, and at least one of an internal combustion engine **210** or an electric drive motor **212**. For example, the automated vehicle **202** can include an autonomous motion technology system **214**. For example, the autonomous motion technology system **214** can include an information acquisition stage **216**, a perception stage **218**, and a control stage **220**.

[0023] For example, the automated vehicle **202** can include an actuator **222** configured to control the steering linkage **206**, one or more of one or more actuators **224** configured to control the one or more brakes **208**, an actuator **226** configured to control a position of a throttle for the internal combustion engine **210** (e.g., if the automated vehicle **202** is capable of being propelled by the internal combustion engine **210**), or an actuator **228** configured to control an amount of current conveyed to the electric drive motor **212** (e.g., if the automated vehicle **202** is capable of being propelled by the electric drive motor **212**).

[0024] For example, the automated vehicle **202** can include one or more of a steering operator interface **230**, a brake operator interface **232**, or an acceleration operator interface **234**. For example, the steering operator interface **230** can include a steering wheel **236**. Alternatively, for example, the steering operator interface **230** can include a joystick-like control lever **238**. For example, the braking operator interface **232** can include a brake pedal **240**. Alternatively, for example, the braking operator interface **232** can include the joystick-like control lever **238**. For example, the accelerating operator interface **234** can include an accelerator pedal **242**. Alternatively, for example, the accelerating operator interface **234** can include the joystick-like control lever **238**.

[0025] For example, the automated vehicle **202** can include one or more of a speedometer **244**, an accelerometer **246**, a gyroscope **248**, a global navigation satellite system (GNSS) **250**, a brake pressure sensor **252**, a wheel speed sensor **254**, a throttle position sensor **256** (e.g., if the automated vehicle **202** is capable of being propelled by the internal combustion engine **210**), an electric drive motor ammeter **258** (e.g., if the automated vehicle **202** is capable of being propelled by the electric drive motor **212**), a steering operator interface position sensor **260**, a steering operator interface applied force sensor **262**, or the like.

[0026] For example, the automated vehicle **202** can include one or more of a lidar device **264**, a radar device **266**, an ultrasonic device **268**, an infrared device **270**, a camera **272**, or the like. For example, the ultrasonic device **268** can be an ultrasonic ranging device, an ultrasonic imaging device, or both. For example, the infrared device **270** can be an infrared ranging device, an infrared imaging device, or both. For example, the camera **272** can be one or more of a color camera, a stereoscopic camera, a video camera, a digital video camera, or the like. For example, using one or more imaging devices (e.g., the camera **272**), a distance and a bearing to an object can be determined using a photogrammetric range imaging technique (e.g., a structure from motion (SfM) technique) applied to a sequence of two-dimensional images.

[0027] For example, the automated vehicle **202** can include a communications device **274**.

[0028] For example, the automated vehicle **202** can include at least a portion **276** of a system **278** for determining a weight to be applied to a rule that governs a change in a movement of an automated vehicle **202**.

[0029] For example, the diagram **200** can include a representation of a longitudinal axis **280** of the automated vehicle **202**.

[0030] With reference to FIGS. **1** and **2**, for example, one or more of: (1) the first vehicle **120**, (2) the second vehicle **122**, or (3) the third vehicle **124** can be the automated vehicle **202**.

[0031] FIG. **3** includes a block diagram that illustrates a first example **300** of a system **302** for determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle, according to the disclosed technologies. For example, the automated vehicle can be the automated vehicle **202** illustrated in FIG. **2**. The first example **300** of the system **302** can be the system **278** illustrated in FIG. **2**. For example, the automated vehicle can be an autonomous vehicle. For example, the rule can be expressed using weighted signal temporal logic. The first example **300** of the system **302** can include a processor **304** and a memory **306**. The memory **306** can be communicably coupled to the processor **304**. For example, the memory **306** can store a weight determination module **308**, a perception module **310**, a rule production module **312**, and an implementation module **314**.

[0032] For example, the weight determination module **308** can include instructions that function to control the processor **304** to determine from responses, from a human subject, to questions about preferences about the manner of the performance of the change of the movement of the automated vehicle, the weight for the rule. For example, the questions can be selected, using a greedy technique, from a question set.

[0033] For example, the perception module **310** can include instructions that function to control the processor **304** to determine an existence of a reason to cause the change of the movement of the automated vehicle.

[0034] For example, the rule production module **312** can include instructions that function to control the processor **304** to produce the rule, with the weight applied, that governs the manner of the performance of the change of the movement of the automated vehicle. For example, the rule can include: (1) a first subrule for a preference with respect to the manner of the performance of the change of the movement of the automated vehicle and (2) a second subrule for a safety of the manner of the performance of the change of the movement of the automated vehicle. For example, the preference can be with respect to one or more of: (1) a degree of comfort of the human subject during the performance of the change of the movement of the automated vehicle, (2) a degree of vehicle performance of the automated vehicle during the performance of the change of the movement of the automated vehicle, or (3) the like.

[0035] For example, the implementation module **314** can include instructions that function to control the processor **304** to cause the change of the movement of the automated vehicle.

[0036] In an implementation, for example, additionally, the memory **306** can further store a presentation module **316** and a reception module **318**. For example, the presentation module **316** can include instructions that function to control the processor **304** to cause the questions to be presented to the human subject. For example, the reception module **318** can include instructions that function to control the processor **304** to cause the processor **304** to receive the responses.

[0037] For example: (1) the instructions to cause the questions to be presented to the human subject, (2) the instructions to receive the responses, and (3) the instructions to determine the weight for the rule can be configured to be performed before the human subject causes the automated vehicle to be operated on a public road. That is, for example, in order to increase the likelihood that the human subject (i.e., the human driver) will accept having the automated vehicle operated under the control of the automated driving system, the weight to be applied to the rule, so that the rule produced by the system **302** can incorporate the preference of the human subject about the manner of the performance of the change of the movement of the automated vehicle, can be determined before the human subject causes the automated vehicle to be operated on a public road.

[0038] Additionally or alternatively, for example, the implementation module **314** can further include instructions that function to control the processor **304** to cause, before a determination of the weight for the rule, the performance of the change of the movement of the automated vehicle in a specific manner. For example: (1) the instructions to cause the questions to be presented to the human subject, (2) the instructions to receive the responses, and (3) the instructions to determine the weight for the rule can be configured to be performed in response to the performance of the change of the movement of the automated vehicle in the specific manner. That is, for example, the automated vehicle can perform the change of the movement of the automated vehicle in the specific manner. For example, after the performance of the change of the movement of the automated vehicle in the specific manner, a question about a preference of the human subject (i.e., the human driver) about the specific manner of the performance of the change of the movement of the automated vehicle can be presented to the human subject (i.e., the human driver), a response can be received, and the weight for the rule can be determined. For example, such a process can be performed in an iterative manner until a final version of the weight for the rule is determined.

[0039] FIG. **4** includes a block diagram that illustrates a second example **400** of the system **302** for determining the weight for the rule that governs the manner of the performance of the change of the movement of the automated vehicle, according to the disclosed technologies. The second example **400** of the system **302** can include, for example, an automated vehicle **402** and a computing device **404**. For example, the computing device **404** can be separate from the automated vehicle **402**. For example, the automated vehicle **402** can be the automated vehicle **202** illustrated in FIG. **2**. For example, the computing device **404** can be the computing device **130** illustrated in FIG. **1**. The second example **400** of the system **302** can include a first processor **406**, a first memory **408**, a second processor **410**, and a second memory **412**. The first memory **408** can be communicably coupled to the first processor **406**. The second memory **412** can be communicably coupled to the second processor **410**. For example, the first processor **406** and the first memory **408** can be disposed on the automated vehicle **402**. For example, the second processor **410** and the second memory **412** can be disposed on the computing device **404**.

[0040] For example, the first memory **408** can store a first portion of the system **302**. In the second example **400** of the system **302**, the first portion of the system **302** can be the portion **276** of the system **278** illustrated in FIG. **2**. For example, the first memory **408** can store the perception module **310**, the rule production module **312**, the implementation module **314**, and an automated vehicle communications module **414**. For example, the instructions to determine the existence of the reason to cause the change of the movement of the automated vehicle can include instructions to cause the first processor **406** to determine the existence of the reason to cause the change of the movement of the automated vehicle. For example, the instructions to produce the rule, with the weight applied, that governs the manner of the performance of the change of the movement of the automated vehicle can include instructions to cause the first processor **406** to produce the rule, with the weight applied that governs the manner of the performance of the change of the movement of the automated vehicle. For example, the instructions to cause the change of the movement of the automated vehicle can include instructions to cause the first processor **406** to cause the change of the movement of the automated vehicle. For example, the automated vehicle communications module **414** can include instructions that function to control the first processor **406** to receive the weight for the rule. With reference to FIG. **2**, for example, the weight for the rule can be received by the communications device **274** included in the automated vehicle **202**.

[0041] For example, the second memory **412** can store a second portion of the system **302**. For example, the second memory **412** can store the weight determination module **308**, the presentation module **316**, the reception module **318**, and a computing device communications module **416**. For example, the instructions to cause the questions to be presented to the human subject can include instructions to cause the second processor **410** to cause the questions to be presented to the human subject. For example, the instructions to receive the responses can include instructions to cause the second processor **410** to receive the responses. For example, the instructions to determine the weight for the rule can include instructions to cause the second processor **410** to determine the weight for the rule. For example, the computing device communications module **416** can include instructions that function to control the second processor **410** to cause the weight for the rule to be transmitted to the automated vehicle **402**. With reference to FIG. **1**, for example, the weight for the rule can be transmitted by the communications device **132** included in the computing device **130**.

[0042] For example, of the questions, a question can include a pairwise comparison question. For example, the pairwise comparison question can include a pair of options. For example, the pair of options can include a first option and a second option. For example, the first option can be about a first manner of the performance of the change of the movement of the automated vehicle. For example, the second option can be about a second manner of the performance of the change of the movement of the automated vehicle. For example, of the responses, a response to the pair of options can include one of the first option and the second option.

[0043] For example, the instructions to cause the pairwise comparison question to be presented to the human subject can include instructions to cause occurrences of: (1) a simulation of the first manner of the performance of the change of the movement of the automated vehicle and (2) a simulation of the second manner of the performance of the change of the movement of the automated vehicle. For example, in the second example **400** of the system **302**, the computing device **404** can be a driving simulator.

[0044] In another implementation, for example, the weight can be a weight valuation from a weight valuations set. For example, a posterior distribution of the weight valuations set can have an initial value that is a quotient of one divided by a count of weight valuations in the weight valuations set. For example, the greedy technique can include performing, in an iterative manner until one or more of: (1) a count of the questions in the question set is zero, (2) a count iterations is greater than a predefined limit of a number of the questions to be presented to the human subject, or (3) the posterior distribution of the weight valuations set is greater than a threshold: (1) determining, from the questions in the question set and based on members of a set of question-answer tuples associated with the question set, a specific question that has a maximum expected information gain over the weight valuations, (2) removing, from the question set, the specific question, (3) receiving a specific response to the specific question, (4) producing an adjustment of the set of question-answer tuples to reflect a selection of a question-answer tuple that includes the specific question and the specific response, (5) producing an update of the posterior distribution of the weight valuations set to reflect the adjustment of the set of question-answer tuples, and (6) determining, based on the update of the posterior distribution of the weight valuations set and for this iteration, a most-likely valuation of the weight from the weight valuations set. For example, the weight valuation can be determined from an intersection of a unit norm ball and a positive quadrant. For example, the maximum expected information gain over the weight valuations can be a function of an entropy of a random variable that represents a weight valuation of the human subject. For example, the maximum expected information gain over the weight valuations can be a function of the predefined limit of the number of the questions to be presented to the human subject. For example, the producing the update of the posterior distribution of the weight valuations set can include producing, using Bayes' Rule, the update of the posterior distribution of the weight valuations set.

[0045] In yet another implementation, for example, the instructions to cause the change of the movement of the automated vehicle can include instructions to cause a first signal to be transmitted to a control stage of an autonomous motion technology system of the automated vehicle. For example, the automated vehicle can be the automated vehicle **202** illustrated in FIG. **2**. For example, the system **302** can be the at least the portion **276** of the system **278** illustrated in FIG. **2**. For example, the control stage can be the control stage **220** illustrated in FIG. **2**. For example, the autonomous motion technology system can be the autonomous motion technology system **214** illustrated in FIG. **2**. For example, the first signal can include information about the rule. For example, the control stage can be configured to: (1) transmit a second signal to an actuator configured to cause the movement of the automated vehicle and (2) receive a third signal from a sensor configured to determine a measurement of a variable associated with the movement of the automated vehicle so that the movement of the automated vehicle occurs in accordance with the rule. For example, the variable can include one or more of a speed of the automated vehicle, an acceleration of the automated vehicle, a jerk of the automated vehicle, a distance between the automated vehicle and an object, a longitudinal distance between a longitudinal axis of the automated vehicle and the object, a lateral distance between the longitudinal axis of the automated vehicle and the object, or the like. For example, in accordance with the rule, the measurement of the variable can be a function of a measurement of another variable.

[0046] Returning to FIG. **1**, for example, the diagram **100** can include a representation of a path **136** between the third vehicle **124** and the pedestrian **134**. A first example of an employment of the disclosed technologies can be a determination of a weight, indicative of a preference of a human subject (i.e., a human driver) of the third vehicle **124**, to be applied to a rule that governs a manner of a performance of a reduction of a speed of the third vehicle **124** along the path **136**.

[0047] FIG. **5** includes graphs **500** that illustrate examples of speed as a function of a specific distance to a point at which an automated vehicle is to stop that can be used for the first example of the employment of the disclosed technologies. For example, the graphs **500** can include a first function **502**, a second function **504**, a third function **506**, a fourth function **508**, a fifth function **510**, a sixth function **512**, a seventh function **514**, an eighth function **516**, and a ninth function **518**. For example, the first function **502** can represent a path in which a rate of the reduction of the speed is constant. For example, the second function **504** can represent a path in which a rate of the reduction of the speed is initially small, but becomes large. For example, the third function **506** can represent a path in which a rate of the reduction of the speed is initially smaller than the path represented by the second function **504**, but becomes larger than the path represented by the second function **504**. For example, the fourth function **508** can represent a path in which a rate of the reduction of the speed is initially smaller than the path represented by the third function **506**, but becomes larger than the path represented by the third function **506**. For example, the fifth function **510** can represent a path in which a rate of the reduction of the speed is initially smaller than the path represented by the fourth function **508**, but becomes larger than the path represented by the fourth function **508**. For example, the sixth function **512** can represent a path in which a rate of the reduction of the speed is initially large, but becomes small. For example, the seventh function **514** can represent a path in which a rate of the reduction of the speed is initially larger than the path represented by the sixth function **512**, but becomes smaller than the path represented by the sixth function **512**. For example, the eighth function **516** can represent a path in which a rate of the reduction of the speed is initially larger than the path represented by the seventh function **514**, but becomes smaller than the path represented by the seventh function **514**. For example, the ninth function **518** can represent a path in which a rate of the reduction of the speed is initially larger than the path represented by the eighth function **516**, but becomes smaller than the path represented by the eighth function **516**.

[0048] With reference to FIGS. **1-5**, for example, of the questions, a question can include a pairwise comparison question. For example, the pairwise comparison question can include a pair of options. For example, the pair of options can include a first option and a second option. For example, the first option can be about a first manner of the performance of the reduction of the speed of the third vehicle **124** along the path **136**. For example, the second option can be about a second manner of the performance of the reduction of the speed of the third vehicle **124** along the path **136**. For example, the presentation module **316** can cause the pairwise comparison question to be presented to the human subject by causing occurrences of: (1) a simulation of the first manner of the performance of the reduction of the speed of the third vehicle **124** along the path **136** and (2) a simulation of the second manner of the performance of the reduction of the speed of the third vehicle **124** along the path **136**.

[0049] For example, in a first iteration, the greedy algorithm can determine that a first pairwise comparison question that includes: (1) as the first option, the performance of the reduction of the speed of the third vehicle **124** along the path **136** represented by the third function **506** and (2) as the second option, the performance of the reduction of the speed of the third vehicle **124** along the path **136** represented by the seventh function **514** has the maximum expected information gain over the weight valuations. For example, in the first iteration, the presentation module **316** can cause: (1) as the first option, a simulation of the performance of the reduction of the speed of the third vehicle **124** along the path **136** represented by the third function **506** and (2) as the second option, a simulation of the performance of the reduction of the speed of the third vehicle **124** along the path **136** represented by the seventh function **514**. For example, in the first iteration, the reception module **318** can receive, as the response to the first pairwise comparison question, an indication that the human subject prefers the simulation of the performance of the reduction of the speed of the third vehicle **124** along the path **136** represented by the third function **506**. For example, in the first iteration, the greedy algorithm can: (1) produce the adjustment of the set of question-answer tuples to reflect a selection of a question-answer tuple that includes the first pairwise comparison question and the response to the first pairwise comparison question, (2) produce the update of the posterior distribution of the weight valuations set to reflect the adjustment of the set of question-answer tuples, and (3) determine, based on the update of the posterior distribution of the weight valuations set and for the first iteration, the most-likely valuation of the weight from the weight valuations set.

[0050] For example, in a second iteration, the greedy algorithm can determine that a second pairwise comparison question that includes: (1) as the first option, the performance of the reduction of the speed of the third vehicle **124** along the path **136** represented by the second function **504** and (2) as the second option, the performance of the reduction of the speed of the third vehicle **124** along the path **136** represented by the fourth function **508** has the maximum expected information gain over the weight valuations. For example, in the second iteration, the presentation module **316** can cause: (1) as the first option, a simulation of the performance of the reduction of the speed of the third vehicle **124** along the path **136** represented by the second function **504** and (2) as the second option, a simulation of the performance of the reduction of the speed of the third vehicle **124** along the path **136** represented by the fourth function **508**. For example, in the second iteration, the reception module **318** can receive, as the response to the second pairwise comparison question, an indication that the human subject prefers the simulation of the performance of the reduction of the speed of the third vehicle **124** along the path **136** represented by the fourth function **508**. For example, in the second iteration, the greedy algorithm can: (1) produce the adjustment of the set of question-answer tuples to reflect a selection of a question-answer tuple that includes the second pairwise comparison question and the response to the second pairwise comparison question, (2) produce the update of the posterior distribution of the weight valuations set to reflect the adjustment of the set of question-answer tuples, and (3) determine, based on the update of the posterior distribution of the weight valuations set and for the second iteration, the most-likely valuation of the weight from the weight valuations set.

[0051] For example, the weight determination module **308** can determine from responses, from the human subject, that the weight to be applied to the rule that governs the manner of the performance of the reduction of the speed of the third vehicle **124** along the path **136** can be associated with the fourth function **508**.

[0052] For example, the perception module **310** of the third vehicle **124** can determine that a presence of one or more of the stop sign **116** or the pedestrian **134** is an existence of a reason to cause the reduction of the speed of the third vehicle **124** along the path **136**.

[0053] For example, the rule production module **312** of the third vehicle **124** can produce the rule, with the weight applied, that governs the manner of the performance of the reduction of the speed of the third vehicle **124** along the path **136**.

[0054] For example, the implementation module **314** of the third vehicle **124** can cause a first signal to be transmitted to the control stage **220** of the autonomous motion technology system **214** of the automated vehicle **202** (i.e., the third vehicle **124**). For example, the control stage **220** can be configured to: (1) transmit a second signal to the one or more actuators **224** configured to control the one or more brakes **208** and (2) receive a third signal from the speedometer **244** so that the movement of the automated vehicle **202** (i.e., the third vehicle **124**) occurs in accordance with the rule.

[0055] Returning to FIG. **1**, for example, the diagram **100** can include a representation of a path **138** between the first vehicle **120** in the eastbound lane **110** and a position in the westbound lane **108** abreast of the second vehicle **122**. A second example of an employment of the disclosed technologies can be a determination of a weight, indicative of a preference of a human subject (i.e., a human driver) of the first vehicle **120**, to be applied to a rule that governs a manner of a performance of a change of a path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122**.

[0056] FIG. **6** includes graphs **600** that illustrate examples of lateral distance as a function of longitudinal distance that can be used for the second example of the employment of the disclosed technologies. For example, the graphs **600** can include a first function **602**, a second function **604**, a third function **606**, a fourth function **608**, a fifth function **610**, a sixth function **612**, a seventh function **614**, an eighth function **616**, and a ninth function **618**. For example, the first function **602** can represent a path in which a change in a lateral direction with respect to a longitudinal direction is constant (i.e., the path **138**). For example, the second function **604** can represent a path in which a change in the lateral direction with respect to the longitudinal direction is initially large, but becomes small. For example, the third function **606** can represent a path in which a change in the lateral direction with respect to the longitudinal direction is initially larger than the path represented by the second function **604**, but becomes smaller than the path represented by the second function **604**. For example, the fourth function **608** can represent a path in which a change in the lateral direction with respect to the longitudinal direction is initially larger than the path represented by the third function **606**, but becomes smaller than the path represented by the third function **606**. For example, the fifth function **610** can represent a path in which a change in the lateral direction with respect to the longitudinal direction is initially larger than the path represented by the fourth function **608**, but becomes smaller than the path represented by the fourth function **608**. For example, the sixth function **612** can represent a path in which a change in the lateral direction with respect to the longitudinal direction is initially small, but becomes large. For example, the seventh function **614** can represent a path in which a change in the lateral direction with respect to the longitudinal direction is initially smaller than the path represented by the sixth function **612**, but becomes larger than the path represented by the sixth function **612**. For example, the eighth function **616** can represent a path in which a change in the lateral direction with respect to the longitudinal direction is initially smaller than the path represented by the seventh function **614**, but becomes larger than the path represented by the seventh function **614**. For example, the ninth function **618** can represent a path in which a change in the lateral direction with respect to the longitudinal direction is initially smaller than the path represented by the eighth function **616**, but becomes larger than the path represented by the eighth function **616**.

[0057] With reference to FIGS. **1-4** and **6**, for example, of the questions, a question can include a pairwise comparison question. For example, the pairwise comparison question can include a pair of options. For example, the pair of options can include a first option and a second option. For example, the first option can be about a first manner of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122**. For example, the second option can be about a second manner of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122**. For example, the presentation module **316** can cause the pairwise comparison question to be presented to the human subject by causing occurrences of: (1) a simulation of the first manner of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** and (2) a simulation of the second manner of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122**.

[0058] For example, in a first iteration, the greedy algorithm can determine that a first pairwise comparison question that includes: (1) as the first option, the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** represented by the third function **606** and (2) as the second option, the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** represented by the seventh function **614** has the maximum expected information gain over the weight valuations. For example, in the first iteration, the presentation module **316** can cause: (1) as the first option. a simulation of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** represented by the third function **606** and (2) as the second option, a simulation of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** represented by the seventh function **614**. For example, in the first iteration, the reception module **318** can receive, as the response to the first pairwise comparison question, an indication that the human subject prefers the simulation of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** represented by the third function **606**. For example, in the first iteration, the greedy algorithm can: (1) produce the adjustment of the set of question-answer tuples to reflect a selection of a question-answer tuple that includes the first pairwise comparison question and the response to the first pairwise comparison question, (2) produce the update of the posterior distribution of the weight valuations set to reflect the adjustment of the set of question-answer tuples, and (3) determine, based on the update of the posterior distribution of the weight valuations set and for the first iteration, the most-likely valuation of the weight from the weight valuations set.

[0059] For example, in a second iteration, the greedy algorithm can determine that a first pairwise comparison question that includes: (1) as the first option, the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** represented by the second function **604** and (2) as the second option, the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** represented by the fourth function **608** has the maximum expected information gain over the weight valuations. For example, in the second iteration, the presentation module **316** can cause: (1) as the first option. a simulation of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** represented by the second function **604** and (2) as the second option, a simulation of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** represented by the fourth function **608**. For example, in the second iteration, the reception module **318** can receive, as the response to the second pairwise comparison question, an indication that the human subject prefers the simulation of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** represented by the second function **604**. For example, in the second iteration, the greedy algorithm can: (1) produce the adjustment of the set of question-answer tuples to reflect a selection of a question-answer tuple that includes the second pairwise comparison question and the response to the second pairwise comparison question, (2) produce the update of the posterior distribution of the weight valuations set to reflect the adjustment of the set of question-answer tuples, and (3) determine, based on the update of the posterior distribution of the weight valuations set and for the second iteration, the most-likely valuation of the weight from the weight valuations set.

[0060] For example, the weight determination module **308** can determine from responses, from the human subject, that the weight to be applied to the rule that governs the manner of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122** can be associated with the second function **604**.

[0061] For example, the perception module **310** of the first vehicle **120** can determine that a presence of the second vehicle **122** is an existence of a reason to cause the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122**.

[0062] For example, the rule production module **312** of the first vehicle **120** can produce the rule, with the weight applied, that governs the manner of the performance of the change of the path of the movement of the first vehicle **120** from the eastbound lane **110** to the westbound lane **108** in order to overtake the second vehicle **122**.

[0063] For example, the implementation module **314** of the first vehicle **120** can cause a first signal to be transmitted to the control stage **220** of the autonomous motion technology system **214** of the automated vehicle **202** (i.e., the first vehicle **120**). For example, the control stage **220** can be configured to: (1) transmit a second signal to the actuator **222** configured to control the steering linkage **206** and (2) receive a third signal from one or more of the lidar device **264**, the radar device **266**, the ultrasonic device **268**, the infrared device **270**, the camera **272**, or the like so that the movement of the automated vehicle **202** (i.e., the first vehicle **120**) occurs in accordance with the rule.

[0064] FIG. **7** includes a flow diagram that illustrates an example of a method **700** that is associated with determining a weight to be applied to a rule that governs a change in a movement of an automated vehicle, according to the disclosed technologies. For example, the automated vehicle can be an autonomous vehicle. For example, the rule can be expressed using weighted signal temporal logic. Although the method **700** is described in combination with the system **302** illustrated in FIGS. **3** and **4**, one of skill in the art understands, in light of the description herein, that the method **700** is not limited to being implemented by the system **302** illustrated in FIGS. **3** and **4**. Rather, the system **302** illustrated in FIGS. **3** and **4** is an example of a system that may be used to implement the method **700**. Additionally, although the method **700** is illustrated as a generally serial process, various aspects of the method **700** may be able to be executed in parallel.

[0065] In FIG. **7**, in the method **700**, at an operation **702**, for example, the weight determination module **308** can determine from responses, from a human subject, to questions about preferences about the manner of the performance of the change of the movement of the automated vehicle, the weight for the rule. For example, the questions can be selected, using a greedy technique, from a question set.

[0066] At an operation **704**, for example, the perception module **310** can determine an existence of a reason to cause the change of the movement of the automated vehicle.

[0067] At an operation **706**, for example, the rule production module **312** can produce the rule, with the weight applied, that governs the manner of the performance of the change of the movement of the automated vehicle. For example, the rule can include: (1) a first subrule for a preference with respect to the manner of the performance of the change of the movement of the automated vehicle and (2) a second subrule for a safety of the manner of the performance of the change of the movement of the automated vehicle. For example, the preference can be with respect to one or more of: (1) a degree of comfort of the human subject during the performance of the change of the movement of the automated vehicle, (2) a degree of vehicle performance of the automated vehicle during the performance of the change of the movement of the automated vehicle, or (3) the like.

[0068] At an operation **708**, for example, the implementation module **314** can cause the change of the movement of the automated vehicle.

[0069] In an implementation, additionally: (1) at an operation **710**, for example, the presentation module **316** can cause the questions to be presented to the human subject and (2) at an operation **712**, for example, the reception module **318** can receive the responses.

[0070] For example: (1) the operation **710**, (2) the operation **712**, and (3) the operation **702** can be performed before the human subject causes the automated vehicle to be operated on a public road.

[0071] Additionally or alternatively, for example, at an operation **714**, the implementation module **314** can cause, before the operation **702**, the performance of the change of the movement of the automated vehicle in a specific manner. For example: (1) the operation **710**, (2) the operation **712**, and (3) the operation **702** can be performed in response to the performance of the change of the movement of the automated vehicle in the specific manner.

[0072] For example, at the operation **710**, the presentation module **316** can be disposed on a computing device **404**. For example, the computing device **404** can be separate from the automated vehicle **402**. For example, at the operation **712**, the reception module **318** can be disposed on the computing device **404**. For example, at the operation **702**, the weight determination module **308** can be disposed on the computing device **404**. Additionally: (1) at an operation **716**, for example, the computing device communications module **416**, disposed on the computing device **404**, can cause the weight for the rule to be transmitted to the automated vehicle **402** and (2) at an operation **718**, for example, the automated vehicle communications module **414**, disposed on the automated vehicle **402**, can receive the weight for the rule. For example, at the operation **704**, the perception module **310** can be disposed on the automated vehicle **402**. For example, at the operation **706**, the rule production module **312** can be disposed on the automated vehicle **402**. For example, at the operation **708**, the implementation module **314** can be disposed on the automated vehicle **402**.

[0073] For example, of the questions, a question can include a pairwise comparison question. For example, the pairwise comparison question can include a pair of options. For example, the pair of options can include a first option and a second option. For example, the first option can be about a first manner of the performance of the change of the movement of the automated vehicle. For example, the second option can be about a second manner of the performance of the change of the movement of the automated vehicle. For example, of the responses, a response to the pair of options can include one of the first option and the second option.

[0074] For example, at the operation **710**, the presentation module **316** can cause occurrences of: (1) a simulation of the first manner of the performance of the change of the movement of the automated vehicle and (2) a simulation of the second manner of the performance of the change of the movement of the automated vehicle. For example, in the second example **400** of the system

302, the computing device **404** can be a driving simulator.

[0075] In another implementation, for example, the weight can be a weight valuation from a weight valuations set. For example, a posterior distribution of the weight valuations set can have an initial value that is a quotient of one divided by a count of weight valuations in the weight valuations set. For example, the greedy technique can include performing, in an iterative manner until one or more of: (1) a count of the questions in the question set is zero, (2) a count iterations is greater than a predefined limit of a number of the questions to be presented to the human subject, or (3) the posterior distribution of the weight valuations set is greater than a threshold: (1) determining, from the questions in the question set and based on members of a set of question-answer tuples associated with the question set, a specific question that has a maximum expected information gain over the weight valuations, (2) removing, from the question set, the specific question, (3) receiving a specific response to the specific question, (4) producing an adjustment of the set of question-answer tuples to reflect a selection of a question-answer tuple that includes the specific question and the specific response, (5) producing an update of the posterior distribution of the weight valuations set to reflect the adjustment of the set of question-answer tuples, and (6) determining, based on the update of the posterior distribution of the weight valuations set and for this iteration, a most-likely valuation of the weight from the weight valuations set. For example, the weight valuation can be determined from an intersection of a unit norm ball and a positive quadrant. For example, the maximum expected information gain over the weight valuations can be a function of an entropy of a random variable that represents a weight valuation of the human subject. For example, the maximum expected information gain over the weight valuations can be a function of the predefined limit of the number of the questions to be presented to the human subject. For example, the producing the update of the posterior distribution of the weight valuations set can include producing, using Bayes' Rule, the update of the posterior distribution of the weight valuations set.

[0076] In yet another implementation, for example, at the operation **708**, the implementation module **314** can cause a first signal to be transmitted to a control stage of an autonomous motion technology system of the automated vehicle. For example, the first signal can include information about the rule. For example, the control stage can be configured to: (1) transmit a second signal to an actuator configured to cause the movement of the automated vehicle and (2) receive a third signal from a sensor configured to determine a measurement of a variable associated with the movement of the automated vehicle so that the movement of the automated vehicle occurs in accordance with the rule. For example, the variable can include one or more of a speed of the automated vehicle, an acceleration of the automated vehicle, a jerk of the automated vehicle, a distance between the automated vehicle and an object, a longitudinal distance between a longitudinal axis of the automated vehicle and the object, a lateral distance between the longitudinal axis of the automated vehicle and the object, or the like. For example, in accordance with the rule, the measurement of the variable can be a function of a measurement of another variable.

[0077] FIG. **8** includes a block diagram that illustrates an example of elements disposed on a vehicle **800**, according to the disclosed technologies. As used herein, a “vehicle” can be any form of powered transport. In one or more implementations, the vehicle **800** can be an automobile. While arrangements described herein are with respect to automobiles, one of skill in the art understands, in light of the description herein, that embodiments are not limited to automobiles. For example, functions and/or operations of one or more of the first vehicle **120** (illustrated in FIG. **1**), the third vehicle **124** (illustrated in FIG. **1**), the automated vehicle **200** (illustrated in FIG. **2**), or the automated vehicle **402** (illustrated in FIG. **4**) can be realized by the vehicle **800**.

[0078] In some embodiments, the vehicle **800** can be configured to switch selectively between an automated mode, one or more semi-automated operational modes, and/or a manual mode. Such switching can be implemented in a suitable manner, now known or later developed. As used herein, “manual mode” can refer that all of or a majority of the navigation and/or maneuvering of the vehicle **800** is performed according to inputs received from a user (e.g., human driver). In one or more arrangements, the vehicle **800** can be a conventional vehicle that is configured to operate in only a manual mode.

[0079] In one or more embodiments, the vehicle **800** can be an automated vehicle. As used herein, “automated vehicle” can refer to a vehicle that operates in an automated mode. As used herein, “automated mode” can refer to navigating and/or maneuvering the vehicle **800** along a travel route using one or more computing systems to control the vehicle **800** with minimal or no input from a human driver. In one or more embodiments, the vehicle **800** can be highly automated or completely automated. In one embodiment, the vehicle **800** can be configured with one or more semi-automated operational modes in which one or more computing systems perform a portion of the navigation and/or maneuvering of the vehicle along a travel route, and a vehicle operator (i.e., driver) provides inputs to the vehicle **800** to perform a portion of the navigation and/or maneuvering of the vehicle **800** along a travel route.

[0080] For example, Standard J3016 202104, Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles, issued by the Society of Automotive Engineers (SAE) International on Jan. 16, 2014, and most recently revised on Apr. 30, 2021,defines six levels of driving automation. These six levels include: (1) level 0, no automation, in which all aspects of dynamic driving tasks are performed by a human driver; (2) level 1, driver assistance, in which a driver assistance system, if selected, can execute, using information about the driving environment, either steering or acceleration/deceleration tasks, but all remaining driving dynamic tasks are performed by a human driver; (3) level 2, partial automation, in which one or more driver assistance systems, if selected, can execute, using information about the driving environment, both steering and acceleration/deceleration tasks, but all remaining driving dynamic tasks are performed by a human driver; (4) level 3, conditional automation, in which an automated driving system, if selected, can execute all aspects of dynamic driving tasks with an expectation that a human driver will respond appropriately to a request to intervene; (5) level 4, high automation, in which an automated driving system, if selected, can execute all aspects of dynamic driving tasks even if a human driver does not respond appropriately to a request to intervene; and (6) level 5, full automation, in which an automated driving system can execute all aspects of dynamic driving tasks under all roadway and environmental conditions that can be managed by a human driver.

[0081] The vehicle **800** can include various elements. The vehicle **800** can have any combination of the various elements illustrated in FIG. **8**. In various embodiments, it may not be necessary for the vehicle **800** to include all of the elements illustrated in FIG. **8**. Furthermore, the vehicle **800** can have elements in addition to those illustrated in FIG. **8**. While the various elements are illustrated in FIG. **8** as being located within the vehicle **800**, one or more of these elements can be located external to the vehicle **800**. Furthermore, the elements illustrated may be physically separated by large distances. For example, as described, one or more components of the disclosed system can be implemented within the vehicle **800** while other components of the system can be implemented within a cloud-computing environment, as described below. For example, the elements can include one or more processors **810**, one or more data stores **815**, a sensor system **820**, an input system **830**, an output system **835**, vehicle systems **840**, one or more actuators **850**, one or more automated driving modules **860**, a communications system **870**, and the system **302** for determining a weight to be applied to a rule that governs a change in a movement of an automated vehicle.

[0082] In one or more arrangements, the one or more processors **810** can be a main processor of the vehicle **800**. For example, the one or more processors **810** can be an electronic control unit (ECU). For example, functions and/or operations of the processor **304** (illustrated in FIG. **3**) or the first processor **406** (illustrated in FIG. **4**) can be realized by the one or more processors **810**.

[0083] The one or more data stores **815** can store, for example, one or more types of data. The one or more data stores **815** can include volatile memory and/or non-volatile memory. Examples of suitable memory for the one or more data stores **815** can include Random-Access Memory (RAM), flash memory, Read-Only Memory (ROM), Programmable Read-Only Memory (PROM), Erasable Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM), registers, magnetic disks, optical disks, hard drives, any other suitable storage medium, or any combination thereof. The one or more data stores **815** can be a component of the one or more processors **810**. Additionally or alternatively, the one or more data stores **815** can be operatively connected to the one or more processors **810** for use thereby. As used herein, “operatively connected” can include direct or indirect connections, including connections without direct physical contact. As used herein, a statement that a component can be “configured to” perform an operation can be understood to mean that the component requires no structural alterations, but merely needs to be placed into an operational state (e.g., be provided with electrical power, have an underlying operating system running, etc.) in order to perform the operation. For example, functions and/or operations of the memory **306** (illustrated in FIG. **3**) or the first memory **408** (illustrated in FIG. **4**) can be realized by the one or more data stores **815**.

[0084] In one or more arrangements, the one or more data stores **815** can store map data **816**. The map data **816** can include maps of one or more geographic areas. In some instances, the map data **816** can include information or data on roads, traffic control devices, road markings, structures, features, and/or landmarks in the one or more geographic areas. The map data **816** can be in any suitable form. In some instances, the map data **816** can include aerial views of an area. In some instances, the map data **816** can include ground views of an area, including 360-degree ground views. The map data **816** can include measurements, dimensions, distances, and/or information for one or more items included in the map data **816** and/or relative to other items included in the map data **816**. The map data **816** can include a digital map with information about road geometry. The map data **816** can be high quality and/or highly detailed.

[0085] In one or more arrangements, the map data **816** can include one or more terrain maps **817**. The one or more terrain maps **817** can include information about the ground, terrain, roads, surfaces, and/or other features of one or more geographic areas. The one or more terrain maps **817** can include elevation data of the one or more geographic areas. The map data **816** can be high quality and/or highly detailed. The one or more terrain maps **817** can define one or more ground surfaces, which can include paved roads, unpaved roads, land, and other things that define a ground surface.

[0086] In one or more arrangements, the map data **816** can include one or more static obstacle maps **818**. The one or more static obstacle maps **818** can include information about one or more static obstacles located within one or more geographic areas. A “static obstacle” can be a physical object whose position does not change (or does not substantially change) over a period of time and/or whose size does not change (or does not substantially change) over a period of time. Examples of static obstacles can include trees, buildings, curbs, fences, railings, medians, utility poles, statues, monuments, signs, benches, furniture, mailboxes, large rocks, and hills. The static obstacles can be objects that extend above ground level. The one or more static obstacles included in the one or more static obstacle maps **818** can have location data, size data, dimension data, material data, and/or other data associated with them. The one or more static obstacle maps **818** can include measurements, dimensions, distances, and/or information for one or more static obstacles. The one or more static obstacle maps **818** can be high quality and/or highly detailed. The one or more static obstacle maps **818** can be updated to reflect changes within a mapped area.

[0087] In one or more arrangements, the one or more data stores **815** can store sensor data **819**. As used herein, “sensor data” can refer to any information about the sensors with which the vehicle **800** can be equipped including the capabilities of and other information about such sensors. The sensor data **819** can relate to one or more sensors of the sensor system **820**. For example, in one or more arrangements, the sensor data **819** can include information about one or more lidar sensors **824** of the sensor system **820**.

[0088] In some arrangements, at least a portion of the map data **816** and/or the sensor data **819** can be located in one or more data stores **815** that are located onboard the vehicle **800**. Additionally or alternatively, at least a portion of the map data **816** and/or the sensor data **819** can be located in one or more data stores **815** that are located remotely from the vehicle **800**.

[0089] The sensor system **820** can include one or more sensors. As used herein, a “sensor” can refer to any device, component, and/or system that can detect and/or sense something. The one or more sensors can be configured to detect and/or sense in real-time. As used herein, the term “real-time” can refer to a level of processing responsiveness that is perceived by a user or system to be sufficiently immediate for a particular process or determination to be made, or that enables the processor to keep pace with some external process.

[0090] In arrangements in which the sensor system **820** includes a plurality of sensors, the sensors can work independently from each other. Alternatively, two or more of the sensors can work in combination with each other. In such a case, the two or more sensors can form a sensor network. The sensor system **820** and/or the one or more sensors can be operatively connected to the one or more processors **810**, the one or more data stores **815**, and/or another element of the vehicle **800** (including any of the elements illustrated in FIG. **8**). The sensor system **820** can acquire data of at least a portion of the external environment of the vehicle **800** (e.g., nearby vehicles). The sensor system **820** can include any suitable type of sensor. Various examples of different types of sensors are described herein. However, one of skill in the art understands that the embodiments are not limited to the particular sensors described herein.

[0091] The sensor system **820** can include one or more vehicle sensors **821**. The one or more vehicle sensors **821** can detect, determine, and/or sense information about the vehicle **800** itself. In one or more arrangements, the one or more vehicle sensors **821** can be configured to detect and/or sense position and orientation changes of the vehicle **800** such as, for example, based on inertial acceleration. In one or more arrangements, the one or more vehicle sensors **821** can include one or more accelerometers, one or more gyroscopes, an inertial measurement unit (IMU), a dead-reckoning system, a global navigation satellite system (GNSS), a global positioning system (GPS), a navigation system **847**, and/or other suitable sensors. The one or more vehicle sensors **821** can be configured to detect and/or sense one or more characteristics of the vehicle **800**. In one or more arrangements, the one or more vehicle sensors **821** can include a speedometer to determine a current speed of the vehicle **800**. For example, functions and/or operations of the speedometer **244** (illustrated in FIG. **2**), the accelerometer **246** (illustrated in FIG. **2**), the gyroscope **248** (illustrated in FIG. **2**), the global navigation satellite system (GNSS) **250** (illustrated in FIG. **2**), the brake pressure sensor **252** (illustrated in FIG. **2**), the wheel speed sensor **254** (illustrated in FIG. **2**), the throttle position sensor **256** (illustrated in FIG. **2**), the electric drive motor ammeter **258** (illustrated in FIG. **2**), the steering operator interface position sensor **260** (illustrated in FIG. **2**), or the steering operator interface applied force sensor **262** (illustrated in FIG. **2**) can be realized by the one or more vehicle sensors **821**.

[0092] Additionally or alternatively, the sensor system **820** can include one or more environment sensors **822** configured to acquire and/or sense driving environment data. As used herein, “driving environment data” can include data or information about the external environment in which a vehicle is located or one or more portions thereof. For example, the one or more environment sensors **822** can be configured to detect, quantify, and/or sense obstacles in at least a portion of the external environment of the vehicle **800** and/or information/data about such obstacles. Such obstacles may be stationary objects and/or dynamic objects. The one or more environment sensors **822** can be configured to detect, measure, quantify, and/or sense other things in the external environment of the vehicle **800** such as, for example, lane markers, signs, traffic lights, traffic signs, lane lines, crosswalks, curbs proximate the vehicle **800**, off-road objects, etc.

[0093] Various examples of sensors of the sensor system **820** are described herein. The example sensors may be part of the one or more vehicle sensors **821** and/or the one or more environment sensors **822**. However, one of skill in the art understands that the embodiments are not limited to the particular sensors described.

[0094] In one or more arrangements, the one or more environment sensors **822** can include one or more radar sensors **823**, one or more lidar sensors **824**, one or more sonar sensors **825**, and/or one more cameras **826**. In one or more arrangements, the one or more cameras **826** can be one or more high dynamic range (HDR) cameras or one or more infrared (IR) cameras. For example, the one or more cameras **826** can be used to record a reality of a state of an item of information that can appear in the digital map. For example, functions and/or operations of the radar device **266** (illustrated in FIG. **2**) can be realized by the one or more radar sensors **823**. For example, functions and/or operations of the lidar device **264** (illustrated in FIG. **2**) can be realized by the one or more lidar sensors **824**. For example, functions and/or operations of the ultrasonic device **268** (illustrated in FIG. **2**) can be realized by the one or more sonar sensors **825**. For example, functions and/or operations of one or more of the infrared device **270** (illustrated in FIG. **2**) or the camera **272** (illustrated in FIG. **2**) can be realized by the one or more cameras **826**.

[0095] The input system **830** can include any device, component, system, element, arrangement, or groups thereof that enable information/data to be entered into a machine. The input system **830** can receive an input from a vehicle passenger (e.g., a driver or a passenger). The output system **835** can include any device, component, system, element, arrangement, or groups thereof that enable information/data to be presented to a vehicle passenger (e.g., a driver or a passenger).

[0096] Various examples of the one or more vehicle systems **840** are illustrated in FIG. **8**. However, one of skill in the art understands that the vehicle **800** can include more, fewer, or different vehicle systems. Although particular vehicle systems can be separately defined, each or any of the systems or portions thereof may be otherwise combined or segregated via hardware and/or software within the vehicle **800**. For example, the one or more vehicle systems **840** can include a propulsion system **841**, a braking system **842**, a steering system **843**, a throttle system **844**, a transmission system **845**, a signaling system **846**, and/or the navigation system **847**. Each of these systems can include one or more devices, components, and/or a combination thereof, now known or later developed. For example, functions and/or operations of the internal combustion engine **210** (illustrated in FIG. **2**) or the electric drive motor **212** (illustrated in FIG. **2**) can be realized by the propulsion system **841**. For example, functions and/or operations of the brakes **208** (illustrated in FIG. **2**) can be realized by the braking system **842**. For example, functions and/or operations of the steering linkage **206** (illustrated in FIG. **2**) can be realized by the steering system **843**.

[0097] The navigation system **847** can include one or more devices, applications, and/or combinations thereof, now known or later developed, configured to determine the geographic location of the vehicle **800** and/or to determine a travel route for the vehicle **800**. The navigation system **847** can include one or more mapping applications to determine a travel route for the vehicle **800**. The navigation system **847** can include a global positioning system, a local positioning system, a geolocation system, and/or a combination thereof.

[0098] The one or more actuators **850** can be any element or combination of elements operable to modify, adjust, and/or alter one or more of the vehicle systems **840** or components thereof responsive to receiving signals or other inputs from the one or more processors **810** and/or the one or more automated driving modules **860**. Any suitable actuator can be used. For example, the one or more actuators **850** can include motors, pneumatic actuators, hydraulic pistons, relays, solenoids, and/or piezoelectric actuators. For example, functions and/or operations of the actuator **222** (illustrated in FIG. **2**), the actuator **224** (illustrated in FIG. **2**), the actuator **226** (illustrated in FIG. **2**), or the actuator **228** (illustrated in FIG. **2**) can be realized by the one or more actuators **850**.

[0099] The one or more processors **810** and/or the one or more automated driving modules **860** can be operatively connected to communicate with the various vehicle systems **840** and/or individual components thereof. For example, the one or more processors **810** and/or the one or more automated driving modules **860** can be in communication to send and/or receive information from the various vehicle systems **840** to control the movement, speed, maneuvering, heading, direction, etc. of the vehicle **800**. The one or more processors **810** and/or the one or more automated driving modules **860** may control some or all of these vehicle systems **840** and, thus, may be partially or fully automated.

[0100] The one or more processors **810** and/or the one or more automated driving modules **860** may be operable to control the navigation and/or maneuvering of the vehicle **800** by controlling one or more of the vehicle systems **840** and/or components thereof. For example, when operating in an automated mode, the one or more processors **810** and/or the one or more automated driving modules **860** can control the direction and/or speed of the vehicle **800**. The one or more processors **810** and/or the one or more automated driving modules **860** can cause the vehicle **800** to accelerate (e.g., by increasing the supply of fuel provided to the engine), decelerate (e.g., by decreasing the supply of fuel to the engine and/or by applying brakes) and/or change direction (e.g., by turning the front two wheels). As used herein, "cause" or "causing" can mean to make, force, compel, direct, command, instruct, and/or enable an event or action to occur or at least be in a state where such event or action may occur, either in a direct or indirect manner.

[0101] The communications system **870** can include one or more receivers **871** and/or one or more transmitters **872**. The communications system **870** can receive and transmit one or more messages through one or more wireless communications channels. For example, the one or more wireless communications channels can be in accordance with the Institute of Electrical and Electronics Engineers (IEEE) 802.11p standard to add wireless access in vehicular environments (WAVE) (the basis for Dedicated Short-Range Communications (DSRC)), the 3rd Generation Partnership Project (3GPP) Long-Term Evolution (LTE) Vehicle-to-Everything (V2X) (LTE-V2X) standard (including the LTE Uu interface between a mobile communication device and an Evolved Node B of the Universal Mobile Telecommunications System), the 3GPP fifth generation (5G) New Radio (NR) Vehicle-to-Everything (V2X) standard (including the 5G NR Uu interface), or the like. For example, the communications system 870 can include "connected vehicle" technology. "Connected vehicle" technology can include, for example, devices to exchange communications between a vehicle and other devices in a packet-switched network. Such other devices can include, for example, another vehicle (e.g., "Vehicle to Vehicle" (V2V) technology), roadside infrastructure (e.g., "Vehicle to Infrastructure" (V2I) technology), a cloud platform (e.g., "Vehicle to Cloud" (V2C) technology), a pedestrian (e.g., "Vehicle to Pedestrian" (V2P) technology), or a network (e.g., "Vehicle to Network" (V2N) technology. "Vehicle to Everything" (V2X) technology can integrate aspects of these individual communications technologies. For example, functions and/or operations of the communications device **126** (illustrated in FIG. **1**), the communications device **128** (illustrated in FIG. **1**), or the communications device **274** (illustrated in FIG. **2**) can be realized by the communications system **870**.

[0102] Moreover, the one or more processors **810**, the one or more data stores **815**, and the communications system **870** can be configured to one or more of form a micro cloud, participate as a member of a micro cloud, or perform a function of a leader of a micro cloud. A micro cloud can be characterized by a distribution, among members of the micro cloud, of one or more of one or more computing resources or one or more data storage resources in order to collaborate on executing operations. The members can include at least connected vehicles.

[0103] The vehicle **800** can include one or more modules, at least some of which are described herein. The modules can be implemented as computer-readable program code that, when executed by the one or more processors **810**, implement one or more of the various processes described herein. One or more of the modules can be a component of the one or more processors **810**. Additionally or alternatively, one or more of the modules can be executed on and/or distributed among other processing systems to which the one or more processors **810** can be operatively connected. The modules can include instructions (e.g., program logic) executable by the one or more processors **810**. Additionally or alternatively, the one or more data store **815** may contain

such instructions.
[0104] In one or more arrangements, one or more of the modules described herein can include artificial or computational intelligence elements, e.g., neural network, fuzzy logic, or other machine learning algorithms. Further, in one or more arrangements, one or more of the modules can be distributed among a plurality of the modules described herein. In one or more arrangements, two or more of the modules described herein can be combined into a single module.
[0105] The vehicle **800** can include one or more automated driving modules **860**. The one or more automated driving modules **860** can be configured to receive data from the sensor system **820** and/or any other type of system capable of capturing information relating to the vehicle **800** and/or the external environment of the vehicle **800**. In one or more arrangements, the one or more automated driving modules **860** can use such data to generate one or more driving scene models. The one or more automated driving modules **860** can determine position and velocity of the vehicle **800**. The one or more automated driving modules **860** can determine the location of obstacles, obstacles, or other environmental features including traffic signs, trees, shrubs, neighboring vehicles, pedestrians, etc.
[0106] The one or more automated driving modules **860** can be configured to receive and/or determine location information for obstacles within the external environment of the vehicle **800** for use by the one or more processors **810** and/or one or more of the modules described herein to estimate position and orientation of the vehicle **800**, vehicle position in global coordinates based on signals from a plurality of satellites, or any other data and/or signals that could be used to determine the current state of the vehicle **800** or determine the position of the vehicle **800** with respect to its environment for use in either creating a map or determining the position of the vehicle **800** in respect to map data.
[0107] The one or more automated driving modules **860** can be configured to determine one or more travel paths, current automated driving maneuvers for the vehicle **800**, future automated driving maneuvers and/or modifications to current automated driving maneuvers based on data acquired by the sensor system **820**, driving scene models, and/or data from any other suitable source such as determinations from the sensor data **819**. As used herein, “driving maneuver” can refer to one or more actions that affect the movement of a vehicle. Examples of driving maneuvers include: accelerating, decelerating, braking, turning, moving in a lateral direction of the vehicle **800**, changing travel lanes, merging into a travel lane, and/or reversing, just to name a few possibilities. The one or more automated driving modules **860** can be configured to implement determined driving maneuvers. The one or more automated driving modules **860** can cause, directly or indirectly, such automated driving maneuvers to be implemented. As used herein, “cause” or “causing” means to make, command, instruct, and/or enable an event or action to occur or at least be in a state where such event or action may occur, either in a direct or indirect manner. The one or more automated driving modules **860** can be configured to execute various vehicle functions and/or to transmit data to, receive data from, interact with, and/or control the vehicle **800** or one or more systems thereof (e.g., one or more of vehicle systems **840**). For example, functions and/or operations of an automotive navigation system can be realized by the one or more automated driving modules **860**.
[0108] Detailed embodiments are disclosed herein. However, one of skill in the art understands, in light of the description herein, that the disclosed embodiments are intended only as examples. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one of skill in the art to variously employ the aspects herein in virtually any appropriately detailed structure. Furthermore, the terms and phrases used herein are not intended to be limiting but rather to provide an understandable description of possible implementations. Various embodiments are illustrated in FIGS. **1-8**, but the embodiments are not limited to the illustrated structure or application.
[0109] The flowchart and block diagrams in the figures illustrate the architecture, functionality, and

operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in flowcharts or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). One of skill in the art understands, in light of the description herein, that, in some alternative implementations, the functions described in a block may occur out of the order depicted by the figures. For example, two blocks depicted in succession may, in fact, be executed substantially concurrently, or the blocks may be executed in the reverse order, depending upon the functionality involved.

[0110] The systems, components and/or processes described above can be realized in hardware or a combination of hardware and software and can be realized in a centralized fashion in one processing system or in a distributed fashion where different elements are spread across several interconnected processing systems. Any kind of processing system or another apparatus adapted for carrying out the methods described herein is suitable. A typical combination of hardware and software can be a processing system with computer-readable program code that, when loaded and executed, controls the processing system such that it carries out the methods described herein. The systems, components, and/or processes also can be embedded in a computer-readable storage, such as a computer program product or other data programs storage device, readable by a machine, tangibly embodying a program of instructions executable by the machine to perform methods and processes described herein. These elements also can be embedded in an application product that comprises all the features enabling the implementation of the methods described herein and that, when loaded in a processing system, is able to carry out these methods.

[0111] Furthermore, arrangements described herein may take the form of a computer program product embodied in one or more computer-readable media having computer-readable program code embodied, e.g., stored, thereon. Any combination of one or more computer-readable media may be utilized. The computer-readable medium may be a computer-readable signal medium or a computer-readable storage medium. As used herein, the phrase “computer-readable storage medium” means a non-transitory storage medium. A computer-readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples of the computer-readable storage medium would include, in a non-exhaustive list, the following: a portable computer diskette, a hard disk drive (HDD), a solid-state drive (SSD), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or flash memory), a portable compact disc read-only memory (CD-ROM), a digital versatile disc (DVD), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. As used herein, a computer-readable storage medium may be any tangible medium that can contain or store a program for use by or in connection with an instruction execution system, apparatus, or device.

[0112] Generally, modules, as used herein, include routines, programs, objects, components, data structures, and so on that perform particular tasks or implement particular data types. In further aspects, a memory generally stores such modules. The memory associated with a module may be a buffer or may be cache embedded within a processor, a random-access memory (RAM), a ROM, a flash memory, or another suitable electronic storage medium. In still further aspects, a module as used herein, may be implemented as an application-specific integrated circuit (ASIC), a hardware component of a system on a chip (SoC), a programmable logic array (PLA), or another suitable hardware component (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a field-programmable gate array (FPGA), or the like) that is embedded with a defined configuration set (e.g., instructions) for performing the disclosed functions.

[0113] Program code embodied on a computer-readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber, cable, radio frequency (RF), etc., or any suitable combination of the foregoing. Computer program code for carrying out operations for aspects of the disclosed technologies may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java™, Smalltalk, C++, or the like, and conventional procedural programming languages such as the "C" programming language or similar programming languages. The program code may execute entirely on a user's computer, partly on a user's computer, as a stand-alone software package, partly on a user's computer and partly on a remote computer, or entirely on a remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0114] The terms "a" and "an," as used herein, are defined as one or more than one. The term "plurality," as used herein, is defined as two or more than two. The term "another," as used herein, is defined as at least a second or more. The terms "including" and/or "having," as used herein, are defined as comprising (i.e., open language). The phrase "at least one of . . . or . . . " as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. For example, the phrase "at least one of A, B, or C" includes A only, B only, C only, or any combination thereof (e.g., AB, AC, BC, or ABC).

[0115] Aspects herein can be embodied in other forms without departing from the spirit or essential attributes thereof. Accordingly, reference should be made to the following claims, rather than to the foregoing specification, as indicating the scope hereof.

Claims

1. A system, comprising: a processor; and a memory storing: a weight determination module including instructions that, when executed by the processor, cause the processor to determine from responses, from a human subject, to questions about preferences about a manner of a performance of a change of a movement of an automated vehicle, a weight for a rule that governs the manner, the questions selected, using a greedy technique, from a question set; a perception module including instructions that, when executed by the processor, cause the processor to determine an existence of a reason to cause the change; a rule production module including instructions that, when executed by the processor, cause the processor to produce the rule with the weight applied, the rule comprising: a first subrule for a preference, and a second subrule for a safety; and an implementation module including instructions that, when executed by the processor, cause the change.

2. The system of claim 1, wherein the preference is with respect to at least one of: a degree of comfort of the human subject during the performance of the change of the movement of the automated vehicle, or a degree of vehicle performance of the automated vehicle during the performance of the change of the movement of the automated vehicle.

3. The system of claim 1, wherein the memory further stores: a presentation module including instructions that, when executed by the processor, cause the processor to cause the questions to be presented to the human subject; and a reception module including instructions that, when executed by the processor, cause the processor to receive the responses.

4. The system of claim 3, wherein: the instructions to cause the questions to be presented to the human subject, the instructions to receive the responses, and the instructions to determine the weight for the rule are configured to be performed before the human subject causes the automated vehicle to be operated on a public road.

5. The system of claim 3, wherein: the implementation module further includes instructions that, when executed by the processor, cause the processor to cause, before a determination of the weight for the rule, the performance of the change of the movement of the automated vehicle in a specific manner, and wherein: the instructions to cause the questions to be presented to the human subject, the instructions to receive the responses, and the instructions to determine the weight for the rule are configured to be performed in response to the performance of the change of the movement of the automated vehicle in the specific manner.

6. The system of claim 3, wherein: the processor comprises a first processor and a second processor, the first processor is disposed on the automated vehicle, the second processor is disposed on a computing device, the computing device being separate from the automated vehicle, the memory comprises a first memory and a second memory, the first memory is disposed on the automated vehicle, the second memory is disposed on the computing device, the instructions to cause the questions to be presented to the human subject include instructions to cause the second processor to cause the questions to be presented to the human subject, the instructions to receive the responses include instructions to cause the second processor to receive the responses, the instructions to determine the weight for the rule include instructions to cause the second processor to determine the weight for the rule, the instructions to determine the existence of the reason to cause the change include instructions to cause the first processor to determine the existence of the reason to cause the change, the instructions to produce the rule with the weight applied include instructions to cause the first processor to produce the rule with the weight applied, the instructions to cause the change of the movement of the automated vehicle include instructions to cause the first processor to cause the change of the movement of the automated vehicle, the second memory stores a computing device communications module including instructions that, when executed by the second processor, cause the second processor to cause the weight for the rule to be transmitted to the automated vehicle, and the first memory stores an automated vehicle communications module including instructions that, when executed by the first processor, cause the first processor to receive the weight for the rule.

7. The system of claim 3, wherein: a question, of the questions, comprises a pairwise comparison question, the pairwise comparison question comprises a pair of options, the pair of options comprises a first option and a second option, the first option is about a first manner of the performance of the change of the movement of the automated vehicle, the second option is about a second manner of the performance of the change of the movement of the automated vehicle, and a response, of the responses, to the pair of options comprises one of the first option and the second option.

8. The system of claim 7, wherein the instructions to cause the pairwise comparison question to be presented to the human subject include instructions to cause occurrences of: a simulation of the first manner of the performance of the change of the movement of the automated vehicle, and a simulation of the second manner of the performance of the change of the movement of the automated vehicle.

9. The system of claim 1, wherein: the weight is a weight valuation from a weight valuations set, a posterior distribution of the weight valuations set has an initial value that is a quotient of one divided by a count of weight valuations in the weight valuations set, the greedy technique comprises performing, in an iterative manner until at least one of: a count of the questions in the question set is zero, a count of iterations is greater than a predefined limit of a number of the questions to be presented to the human subject, or the posterior distribution of the weight valuations set is greater than a threshold: determining, from the questions in the question set and based on members of a set of question-answer tuples associated with the question set, a specific question that has a maximum expected information gain over the weight valuations, removing, from the question set, the specific question, receiving a specific response to the specific question, producing an adjustment of the set of question-answer tuples to reflect a selection of a question-answer tuple that includes the specific question and the specific response, producing an update of the posterior distribution of the weight valuations set to reflect the adjustment of the set of question-answer tuples, and determining, based on the update of the posterior distribution of the weight valuations set and for this iteration, a most-likely valuation of the weight from the weight

valuations set.
10. The system of claim 9, wherein the weight valuation is determined from an intersection of a unit norm ball and a positive quadrant.
11. The system of claim 9, wherein the maximum expected information gain over the weight valuations is a function of an entropy of a random variable that represents a weight valuation of the human subject.
12. The system of claim 9, wherein the maximum expected information gain over the weight valuations is a function of the predefined limit of the number of the questions to be presented to the human subject.
13. The system of claim 9, wherein the producing the update of the posterior distribution of the weight valuations set comprises producing, using Bayes' Rule, the update of the posterior distribution of the weight valuations set.
14. The system of claim 1, wherein: the instructions to cause the change of the movement of the automated vehicle include instructions to cause a first signal to be transmitted to a control stage of an autonomous motion technology system of the automated vehicle, the first signal includes information about the rule, and the control stage is configured to: transmit a second signal to an actuator configured to cause the movement of the automated vehicle, and receive a third signal from a sensor configured to determine a measurement of a variable associated with the movement of the automated vehicle so that the movement of the automated vehicle occurs in accordance with the rule.
15. The system of claim 14, wherein the variable comprises at least one of a speed of the automated vehicle, an acceleration of the automated vehicle, a jerk of the automated vehicle, a distance between the automated vehicle and an object, a longitudinal distance between a longitudinal axis of the automated vehicle and the object, or a lateral distance between the longitudinal axis of the automated vehicle and the object.
16. The system of claim 14, wherein, in accordance with the rule, the measurement of the variable is a function of a measurement of another variable.
17. A method, comprising: determining, by a processor and from responses, from a human subject, to questions about preferences about a manner of a performance of a change of a movement of an automated vehicle, a weight for a rule that governs the manner, the questions selected, using a greedy technique, from a question set; determining, by the processor, an existence of a reason to cause the change; producing, by the processor, the rule with the weight applied, the rule comprising: a first subrule for a preference, and a second subrule for a safety; and causing, by the processor, the change.
18. The method of claim 17, wherein the automated vehicle is an autonomous vehicle.
19. The method of claim 17, wherein the rule is expressed using weighted signal temporal logic.
20. A non-transitory computer-readable medium for determining a weight for a rule that governs a manner of a performance of a change of a movement of an automated vehicle, the non-transitory computer-readable medium including instructions that, when executed by one or more processors, cause the one or more processors to: determine, from responses, from a human subject, to questions about preferences about the manner of the performance of the change of the movement of the automated vehicle, the weight the weight for the rule that governs the manner of the performance of the change of the movement of the automated vehicle, the questions selected, using a greedy technique, from a question set; determine an existence of a reason to cause the change of the movement of the automated vehicle; produce, the rule, with the weight applied, that governs the manner of the performance of the change of the movement of the automated vehicle, the rule comprising: a first subrule for a preference with respect to the manner of the performance of the change of the movement of the automated vehicle, and a second subrule for a safety of the manner of the performance of the change of the movement of the automated vehicle; and cause the change of the movement of the automated vehicle.